

## DNA Damage and Repair

If the genome is the 'instruction manual' for building and operating the organism, then one can reasonably ask, how does it guard against damage - cosmic 'typos'? Actually it is estimated that the rate of random damage is 10,000 hits per cell per day. So all organisms are able to sense and repair damaged DNA. Sometimes the mistakes are not found, sometimes the repairs are wrong. Of those 10,000, on average one to them will not be fixed, leading to a mutation. So think about the difference between the use of information in the instruction manual for a living thing vs say, the source code for building a computer program, or the blueprints of a space shuttle. In comparison, we are amazingly resistant to changes in the manual.

With respect to repair remember also that the information stored in DNA is present in multiple copies in the cell. Because of the base-pairing rules, damage to one strand may be repaired by reading the other. If that is insufficient, except for gametes - and males, the information is stored on two copies of the chromosomes. Between S phase and the M phase of the cell cycle, the information may be present in even more copies.

### Reading

**MBoC(6th) Ch5:** *DNA REPAIR, HOMOLOGOUS RECOMBINATION (first part)*, pgs 266-280. **Ch20:** *Cancer: pgs 1108-1111, 1116-1119.*

This material is fairly concisely laid out in the reading. With respect to homologous recombination, focus on just the first part of the reading; we will cover the mechanisms more carefully in the module on Recombination.

Pay attention to the section on '*The chemistry of the DNA bases ...*'. There are a multitude of different bases that living things could have chosen to use in DNA. A, C, G, and T have a really important property; if you damage them, they do not turn into each other. The only exception is the one you just met studying epigenetics, <sup>5m</sup>C.

### Need to know and understand:

Errors during replication: Mis-incorporation, strand slippage, over/under replication ( $2N \Rightarrow 4N$ )

Errors during mitosis: Failure to exactly divide ( $4N \Rightarrow 2N+2N$ )

Damage:

- Adduct addition (adding something to a nucleotide that shouldn't be there, e.g. Methylation by SAM)
- Hydrolysis (Taking something away that should be there) e.g. deamination, depurination.
- Oxidation (often breaks)
- Double stranded break - ionizing radiation.
- Interstrand cross-linking. (This is the nastiest of the lot) Usually chemical, e.g. NO, chemotherapeutics
- Intrastrand cross-linking. e.g. UV induced thymidine dimers.

## Repair mechanisms:

- Proof-reading during DNA replication
- Focal. These repair some of the most common (e.g. thymidine dimers by Photolyase) and fix nothing else.
- Base Excision Repair (BER)
- Nucleotide Excision Repair (NER). Often coupled to transcription.
- Mismatch Repair (MMR). Very active during DNA replication
- SOS repair and translesion polymerases (TLS)
- Homologous recombination. Very active during DNA replication.
- Non-Homologous End Joining

DNA repair in Medicine. With the previous processes we covered there was almost no chance of seeing a patient deficient in any part of the basic machinery. With DNA repair this is no longer so. There are two reasons for this. First, unlike the other processes, repair is not part of the basic machinery that handles information in the cell, so a lack does not automatically imply death. Secondly, almost all types of DNA damage can be fixed by more than one mechanism. If one pathway is missing, another can compensate. However, this has consequences. The consequences of partial failure of the DNA repair machinery is manifested in a set of genetic predispositions to cancer, and is associated with accumulating repair failure during tumor progression.

Focus illustrations: Source: Molecular Biology of the Cell: Sixth Edition (2015) Alberts et al., Garland Science, NY

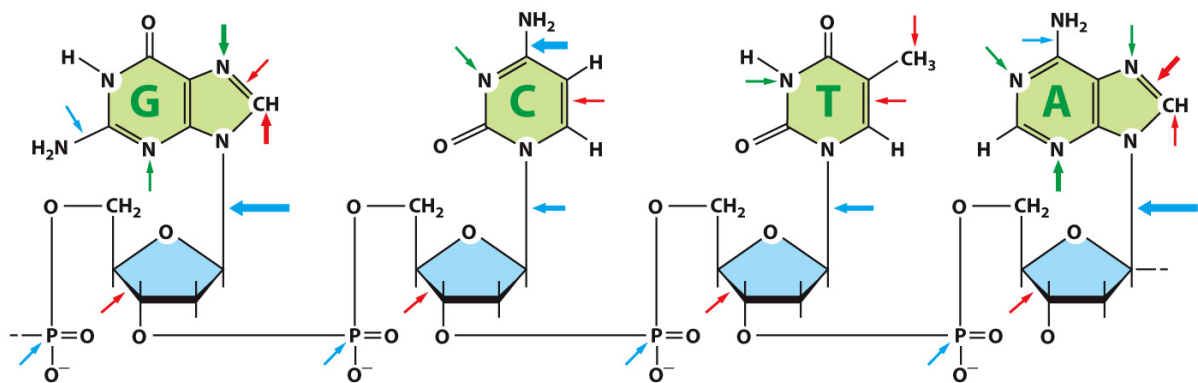


Figure 5-37 Molecular Biology of the Cell 6e (© Garland Science 2015)

**Figure 5–37** A summary of spontaneous alterations that require DNA repair.

The sites on each nucleotide modified by spontaneous oxidative damage (*red arrows*), hydrolytic attack (*blue arrows*), and methylation (*green arrows*) are shown, with the width of each arrow indicating the relative frequency of each event (see Table 5–3). (After T. Lindahl, *Nature* 362:709–715, 1993).

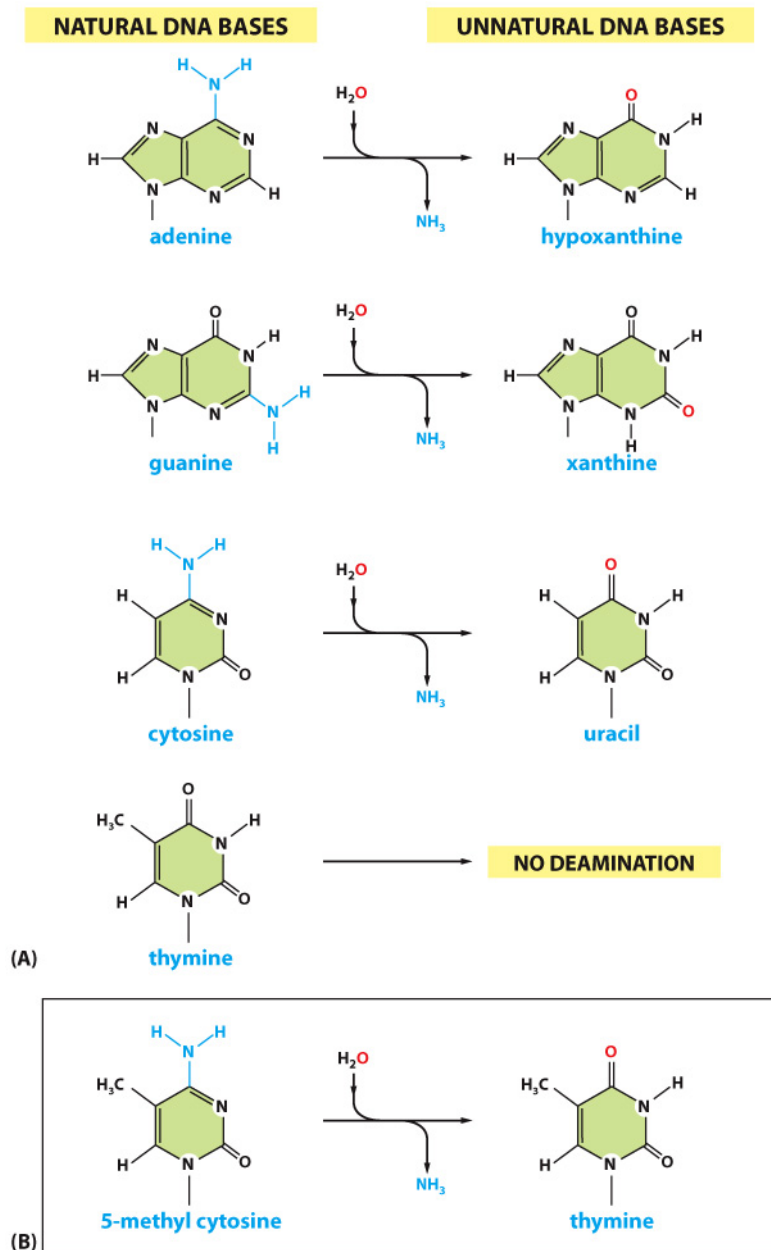


Figure 5-43 Molecular Biology of the Cell 6e (© Garland Science 2015)

**Figure 5-43** The deamination of DNA nucleotides. In each case, the oxygen atom that is added in this reaction with water is colored *red*. (A) The spontaneous deamination products of A and G are recognizable as unnatural when they occur in DNA and thus are readily found and repaired. The deamination of C to U was also illustrated in Figure 5-38; T has no amino group to remove. (B) About 3% of the C nucleotides in vertebrate DNAs are methylated to help in controlling gene expression (discussed in Chapter 7). When these 5-methyl C nucleotides are accidentally deaminated, they form the natural nucleotide T. However, this T will be paired with a G on the opposite strand, forming a mismatched base pair.

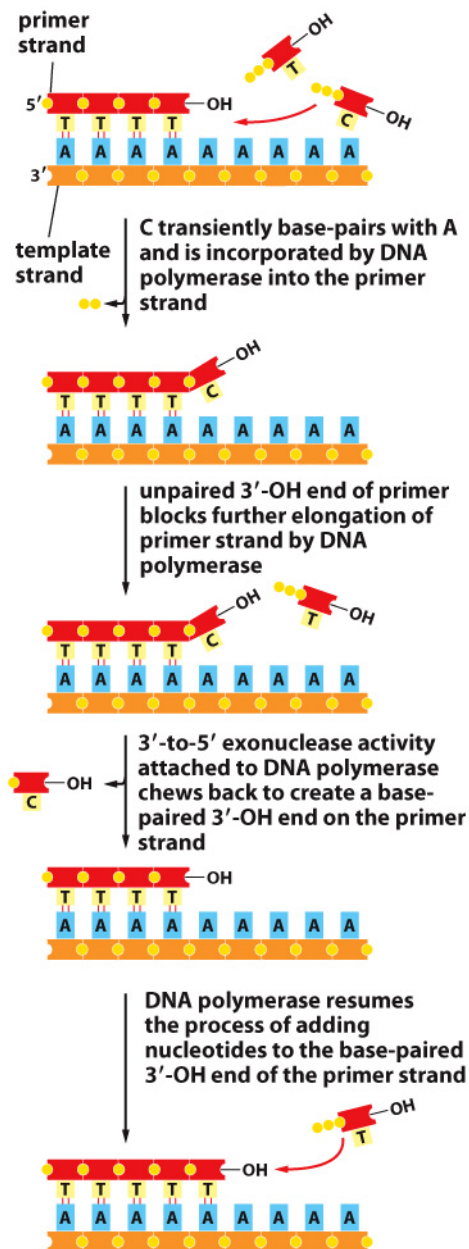


Figure 5-8 Molecular Biology of the Cell 6e (© Garland Science 2015)

**Figure 5–8** Exonucleolytic proofreading by DNA polymerase during DNA replication. In this example, a C is accidentally incorporated at the growing 3'-OH end of a DNA chain. The part of DNA polymerase that removes the misincorporated nucleotide is a specialized member of a large class of enzymes, known as *exonucleases*, that cleave nucleotides one at a time from the ends of polynucleotides.

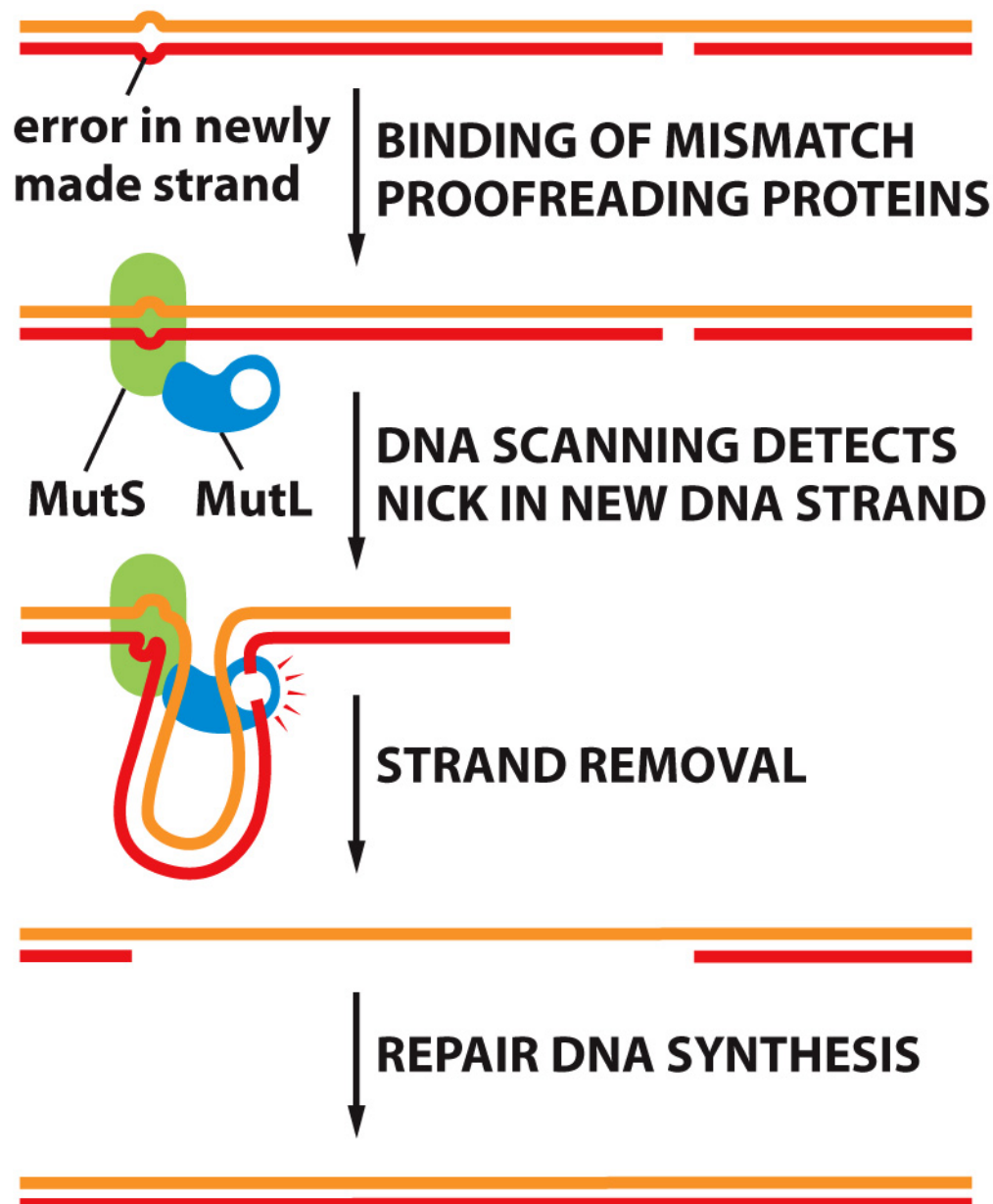


Figure 5-19a Molecular Biology of the Cell 6e (© Garland Science 2015)

**Figure 5–19** Strand-directed mismatch repair. (A) The two proteins shown are present in both bacteria and eukaryotic cells: MutS binds specifically to a mismatched base pair, while MutL scans the nearby DNA for a nick. Once MutL finds a nick, it triggers the degradation of the nicked strand all the way back through the mismatch. Because nicks are largely confined to newly replicated strands in eukaryotes, replication errors are selectively removed. In bacteria, an additional protein in the complex (MutH) nicks unmethylated (and therefore newly replicated) GATC sequences, thereby beginning the process illustrated here. In eukaryotes, MutL contains a DNA nicking activity that aids in the removal of the damaged strand.

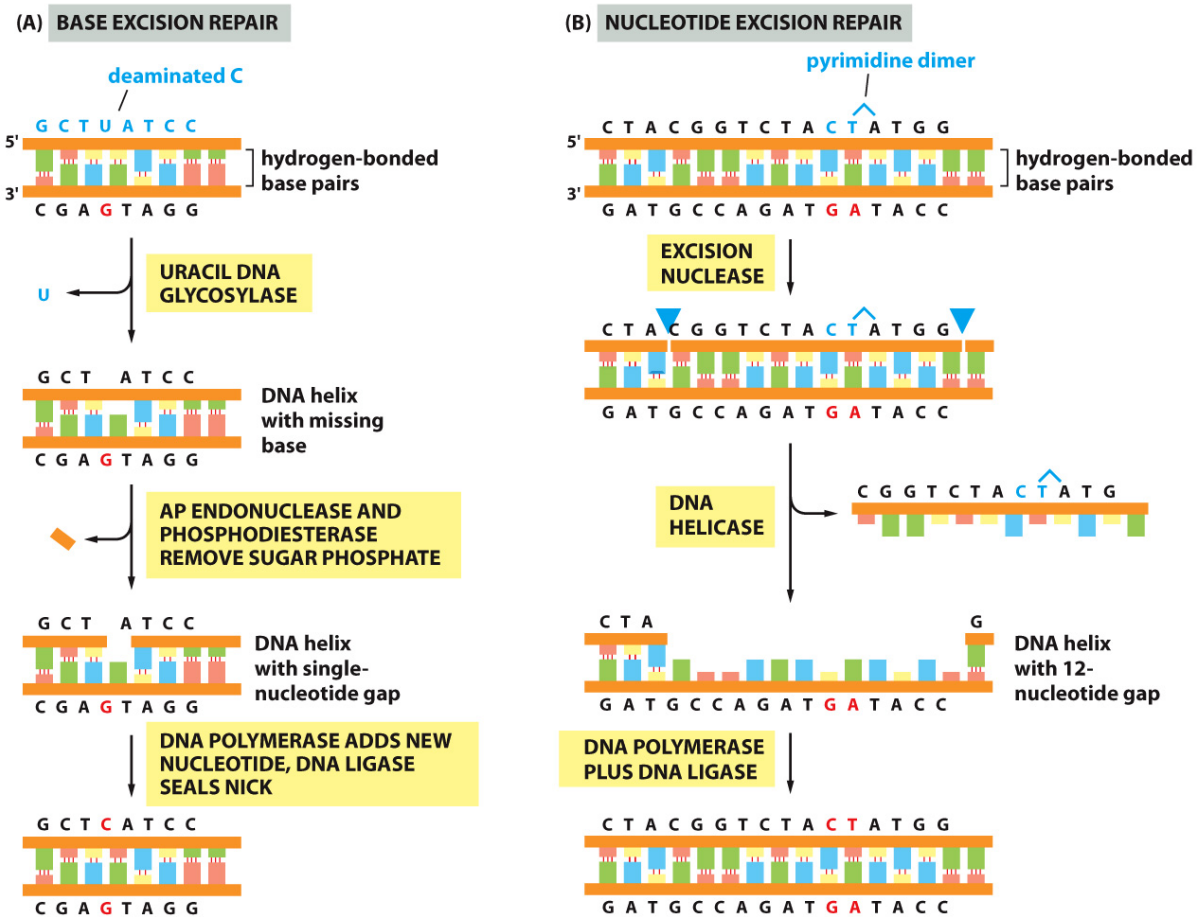


Figure 5-41 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 5– 41 A comparison of two major DNA repair pathways. (A) Base excision repair. This pathway starts with a DNA glycosylase. Here, the enzyme uracil DNA glycosylase removes an accidentally deaminated cytosine in DNA. After the action of this glycosylase (or another DNA glycosylase that recognizes a different kind of damage), the sugar phosphate with the missing base is cut out by the sequential action of AP endonuclease and a phosphodiesterase. (These same enzymes begin the repair of depurinated sites directly.) The gap of a single nucleotide is then filled by DNA polymerase and DNA ligase. The net result is that the U that was created by accidental deamination is restored to a C. AP endonuclease is so-named because it recognizes any site in the DNA helix that contains a deoxyribose sugar with a missing base; such sites can arise either by the loss of a purine (apurinic sites) or by the loss of a pyrimidine (apyrimidinic sites). (B) Nucleotide excision repair. In bacteria, after a multienzyme complex has recognized a lesion such as a pyrimidine dimer (see Figure 5–39), one cut is made on each side of the lesion, and an associated DNA helicase then removes the entire portion of the damaged strand. The excision repair machinery in bacteria leaves the gap of 12 nucleotides shown. In humans, once the damaged DNA is recognized, a helicase is recruited to unwind the DNA duplex locally. Next, the excision nuclease enters and cleaves on either side of the damage, leaving a gap of about 30 nucleotides. The nucleotide excision repair machinery in both bacteria and humans can recognize and repair many different types of DNA damage.

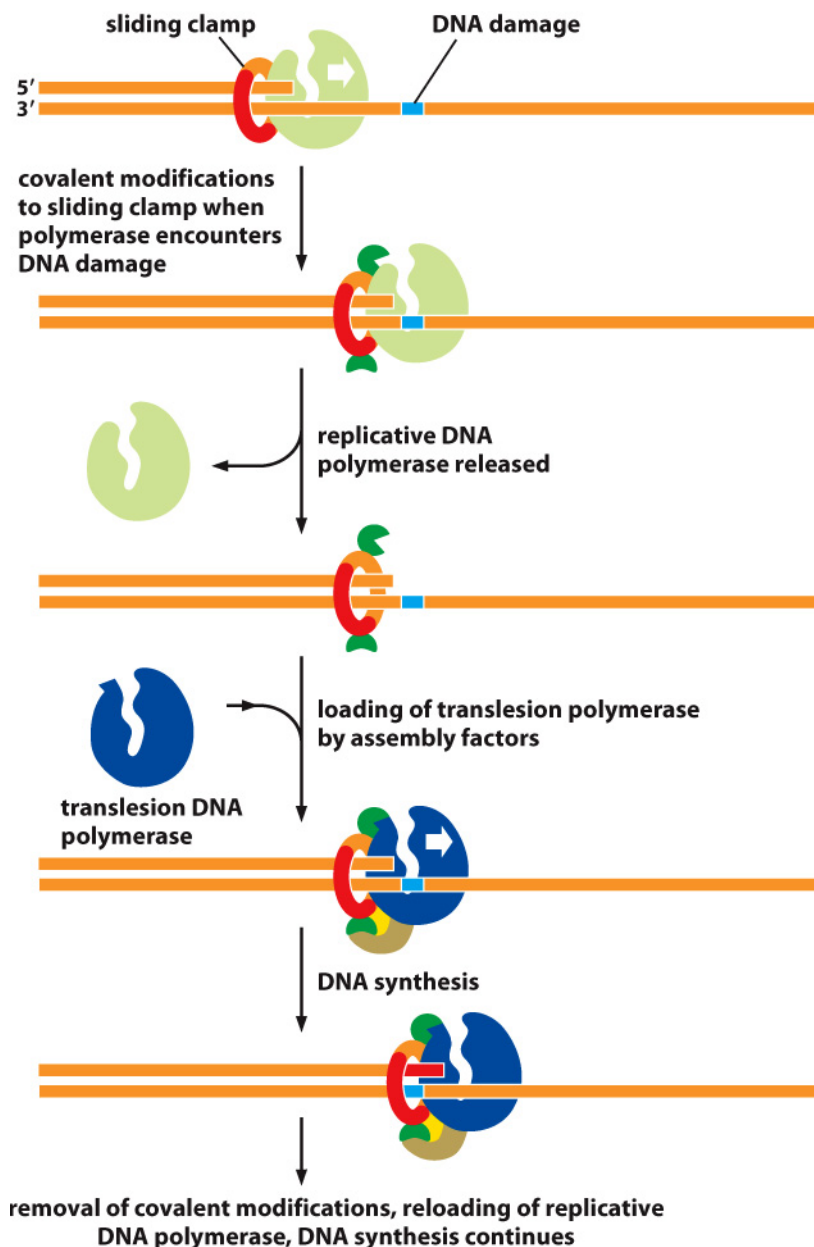


Figure 5-44 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 5–44 Translesion DNA polymerases can use damaged templates. According to this model, a replicative polymerase stalled at a site of DNA damage is recognized by the cell as needing rescue. Specialized enzymes covalently modify the sliding clamp (typically, it is ubiquitylated—see Figure 3–69) which releases the replicative DNA polymerase and, together with damaged DNA, attracts a translesion polymerase specific to that type of damage. Once the damaged DNA is bypassed, the covalent modification of the clamp is removed, the translesion polymerase dissociates, and the replicative polymerase is brought back into play.

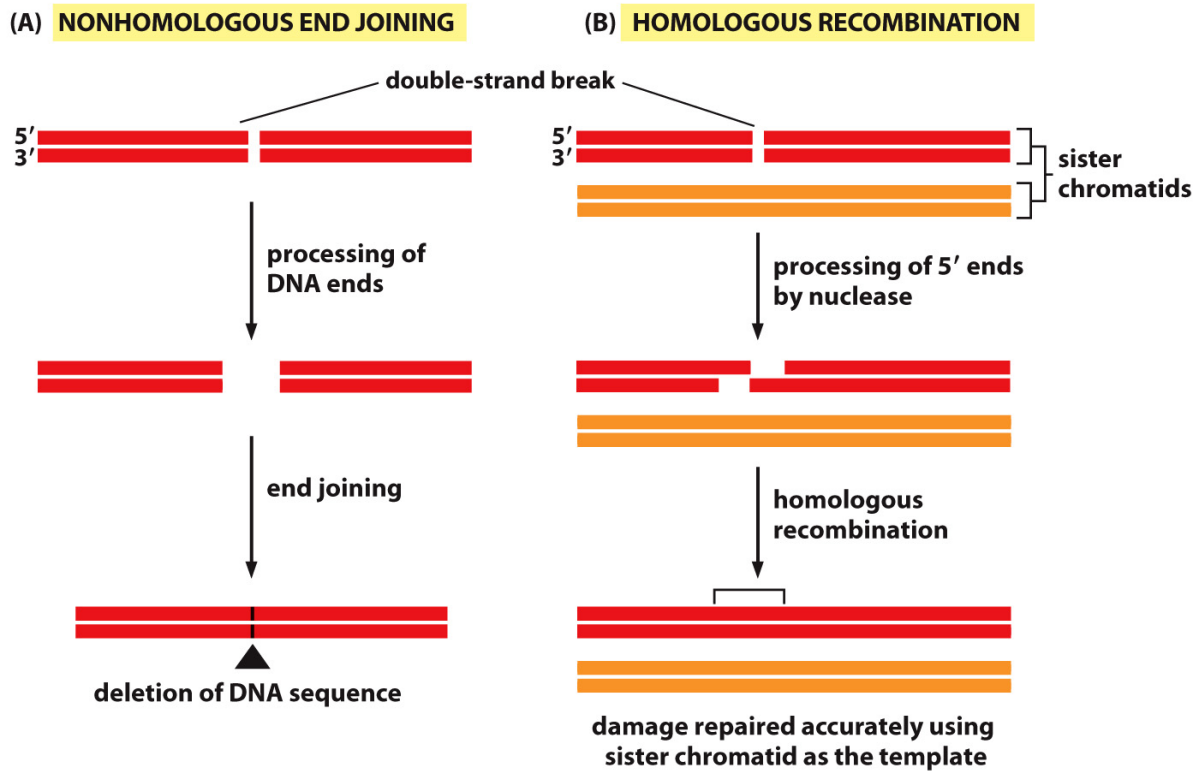


Figure 5-45 Molecular Biology of the Cell 6e (© Garland Science 2015)

Figure 5–45 Two ways to repair double-strand breaks. (A) Nonhomologous end joining alters the original DNA sequence when repairing a broken chromosome. The initial degradation of the broken DNA ends is important because the nucleotides at the site of the initial break are often damaged and cannot be ligated. Nonhomologous end joining usually takes place when cells have not yet duplicated their DNA. (B) Repairing double-strand breaks by homologous recombination is more difficult to accomplish but restores the original DNA sequence. It typically takes place after the DNA has been duplicated (when a duplex template is available) but before the cell has divided. Details of the homologous recombination pathway are presented in the following section (see Figure 5–48).